HR ANALYTICS DASHBOARD

Final Project Report

Project Domain	Human Resources / Data Analytics
Tools Used	Microsoft Power BI, HR dataset, GitHub for project sharing
Prepared By	Project Team / Student
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This report documents the problem, dataset characteristics visible in the dashboard, data preparation approach, analysis, dashboard design, and business insights. Details that require the original dataset are identified for verification rather than assumed.

1. Executive Summary

This project uses an interactive Power BI dashboard to summarize workforce information and support HR reporting. The dashboard presents employee headcount, average age, average years of experience, annual salary, department distribution, salary slabs, age groups, gender distribution, job-role satisfaction, and employee experience patterns. These views help a reader explore workforce composition and identify areas that may need further investigation.

The screenshot shows approximately 1.417K employees, an average age of 36.94 years, average years of experience of 7.04, and an annual salary measure displayed as 957M. The meaning and currency/unit of the salary figure should be confirmed against the underlying dataset and Power BI measure before external publication.

2. Problem Statement

HR teams need a clear and consistent way to understand workforce composition, employee experience, salary distribution, department size, and satisfaction. When these indicators are spread across tables or separate reports, it can be difficult to identify patterns quickly. The objective of this project is to consolidate key HR metrics into a single interactive dashboard so users can review workforce data and support evidence-based planning.

3. Project Objectives

- Present headline workforce KPIs in one place.
- Compare employee counts across departments and age groups.
- Explore salary slabs and gender distribution.
- Review job-role satisfaction ratings shown in the report.
- Visualize the distribution of employees by years of experience.
- Summarize observations that HR teams can validate and investigate further.

4. Dataset Information

The dashboard appears to use an HR employee dataset with fields or derived measures for employee records, age, years of experience, annual salary, department, salary slab, gender, job role, and job satisfaction. The screenshot shows a total of 1,417 employee records. The exact dataset name, source URL, number of raw columns, date coverage, and licensing cannot be established from the screenshot alone and should be filled in from the original CSV/XLSX file.

Field / Measure	Purpose in Dashboard
Employee count	Overall workforce size and department counts
Age	Average age and age-group distribution
Years of experience	Experience distribution and average experience
Annual salary / Salary slab	Salary summary and salary-band comparison
Department	Compare workforce size by department
Gender	Show gender distribution
Job role / Job satisfaction	Compare satisfaction ratings by role

5. Data Cleaning and Preparation

The following checks should be performed on the source file and documented with the actual actions taken. The screenshot does not prove which cleaning operations were applied, so these items are a reproducible checklist rather than a claim that every step was completed.

1. Inspect column names and confirm that each field has a clear meaning.
2. Check missing values in age, department, gender, job role, salary, experience, and satisfaction fields.
3. Check duplicate employee records; remove duplicates only when they are true duplicates and the chosen key supports that decision.
4. Set correct data types: numeric for age, experience and salary; categorical/text for department, gender, role and salary slab.
5. Standardize inconsistent category labels and trim extra spaces.
6. Validate age, experience and salary values for impossible or suspicious entries; investigate before deleting or changing them.
7. Confirm that salary slabs match the salary values and that satisfaction ratings use a consistent scale.
8. Refresh the Power BI model and verify that the KPI totals match the prepared dataset.

Recommended documentation: record the number of rows before and after cleaning, the number of duplicates handled, missing-value treatment, and any rows excluded. Do not report counts unless measured from the actual data.

6. Analysis and Dashboard Components

The supplied screenshot shows the following visuals:

Headline KPIs: Total Employee = 1.417K; Average of Age = 36.94; Average of Years of Experience = 7.04; Annual Salary measure = 957M.

Employee by Department: Operations is the largest group shown (512; 36.13%), followed by Administration (406; 28.65%) and Sales (207; approximately 14.6%). Other displayed departments include IT (166; 11.71%), Marketing (64), Human Resources (39), and Finance (23). The department counts shown add to 1,417.

Salary Slab: The chart compares employee counts across 0–3 LPA, 3–6 LPA, 6–10 LPA, and 10+ LPA bands, with a legend for Yes/No. The exact meaning of the Yes/No series should be checked in the model before interpreting it.

Age Group: The largest displayed group is 26–35, followed by 36–45. The 55+ group is the smallest. Exact values should be confirmed from the visual or dataset.

Gender Distribution: The chart displays 800 employees (56.46%) in one gender category and 617 (43.54%) in the other. The legend identifies the categories as Male and Female; the screenshot's color-to-category mapping should be verified before describing which number belongs to each category.

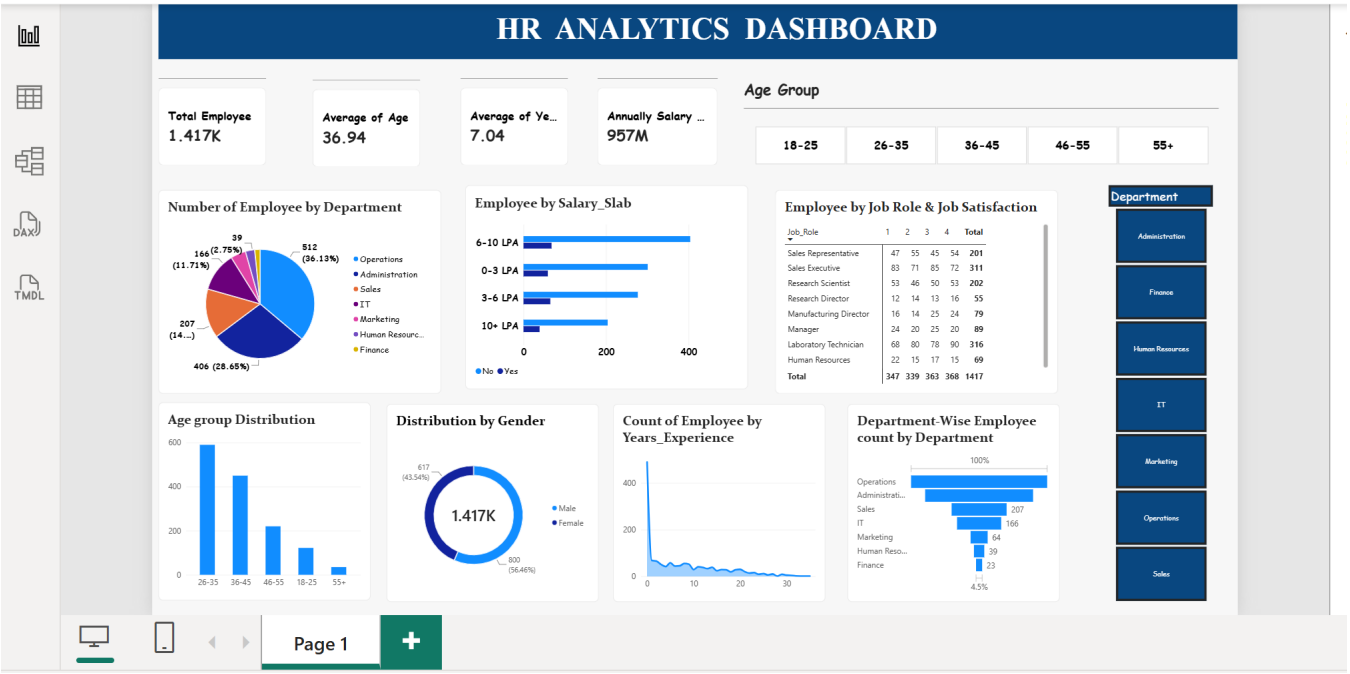
Job Role and Job Satisfaction: A matrix displays ratings 1–4 and totals by role. It can be used to compare satisfaction-rating counts across job roles, but the underlying survey definition and rating scale should be confirmed.

Years of Experience: The line/area visual shows a high count at the lowest experience value and decreasing counts at higher values. Confirm the axis field and bins before drawing conclusions about the exact distribution.

Department filter: The right-side department slicer allows the user to filter dashboard visuals by department. Age-group buttons provide another interactive filter.

7. Dashboard Screenshot

The following screenshot was supplied for this report. It is included as a visual reference of the dashboard layout.



8. Key Business Insights

1. Workforce is concentrated in two departments. Operations and Administration account for 512 + 406 = 918 employees, approximately 64.8% of the 1,417 employees shown. Workforce planning and workload review may therefore benefit from particular attention to these two departments.

2. The workforce is mainly in the 26–45 age range. The age chart shows the largest groups in 26–35 and 36–45. HR may review career progression, learning needs and succession planning for these groups; exact percentages should be calculated from the dataset.

3. The dashboard includes a meaningful experience profile. Average years of experience is displayed as 7.04. This can support planning for onboarding, development and role progression, once the measure's definition is confirmed.

4. Salary bands can support compensation review. The salary-slab visual allows HR to compare employee counts across salary ranges. A follow-up analysis should validate whether each band is aligned with role, department, experience and internal pay-equity policies.

5. Satisfaction varies by job role and rating. The matrix provides a starting point for identifying roles with relatively more low or high satisfaction ratings. HR should inspect rating distributions and sample sizes before deciding on interventions.

6. Department and age filters support targeted review. The slicers let users narrow the view to specific groups rather than relying only on overall totals. This can help HR compare patterns across departments and age bands.

These are observations from the visible dashboard and suggested uses of the data, not proof of causes. The report does not infer why a pattern exists; follow-up analysis is required before policy decisions.

9. Recommendations

• Validate the salary total and currency/unit used by the annual salary measure.
• Confirm the meaning of the Yes/No series in the salary chart.
• Review employee satisfaction by job role and investigate roles with comparatively low satisfaction.
• Compare department size with workload, staffing plans and business needs.
• Maintain a documented cleaning log and reconcile dashboard totals with the source dataset.
• Protect employee privacy: share aggregated results and avoid publishing personal or identifiable employee records.

10. Limitations

This report is based on one dashboard screenshot, not a full inspection of the source dataset or Power BI model. As a result, the dataset source, column count, cleaning results, exact age-band counts, satisfaction scale interpretation, salary currency/unit, and some chart mappings require verification. The dashboard is descriptive and does not establish causation.

11. Conclusion

The HR Analytics Dashboard consolidates key workforce indicators into one interactive Power BI report. It provides a useful overview of headcount, departments, age, experience, salary bands, gender and job-role satisfaction. With source-data validation and documented cleaning steps, the dashboard can support more consistent HR reporting and help identify areas for further investigation.

12. Project Files and Submission Checklist

Before submitting to GitHub, include: (1) this report, (2) the Power BI .pbix file, (3) original and cleaned dataset if permitted, (4) dashboard screenshot, and (5) a README.md with the project overview and tools. Replace placeholders such as dataset source and prepared-by details with verified information. If the dataset contains private employee data, upload only an anonymized or approved version.